

Daily Problem: Validity of a Sorting Network is coNP-Complete

Problem Statement

Prove that the problem of checking whether a given comparator network is a valid sorting network is coNP-complete.

Solution

Let VALID be the language of comparator networks that sort every input sequence. By the zero-one principle, a comparator network is a sorting network if and only if it sorts every binary input. So it is enough to reason about 0-1 inputs.

Step 1: VALID belongs to coNP

Consider the complement language

$$\text{NONSORT} = \{ N \mid \text{there exists a 0-1 input } x \text{ such that } N(x) \text{ is not sorted} \}.$$

A certificate for NONSORT is simply the bad input x . Given N and x , we simulate the network on x and then check whether the output is sorted. Both tasks take polynomial time. Hence NONSORT belongs to NP. Therefore VALID belongs to coNP.

Step 2: VALID is coNP-hard

To prove coNP-hardness of VALID, it is enough to prove NP-hardness of its complement NONSORT. This is a classical theorem: Chung and Ravikumar proved that deciding whether a given comparator network fails on some 0-1 input is NP-complete. Parberry later cites this result explicitly in his paper on sorting-network verification and uses it as the starting point for stronger restricted-depth hardness results.

Since NONSORT is NP-hard, its complement VALID is coNP-hard.

Step 3: Conclude coNP-completeness

We have shown that VALID belongs to coNP and that VALID is coNP-hard. Hence VALID is coNP-complete.

Conclusion

The problem of checking whether a comparator network is a valid sorting network is coNP-complete. The easy half is membership in coNP, using the zero-one principle and a bad binary input as certificate. The hard half follows from the classical NP-completeness of the complement problem NONSORT.

Source Note

Hardness is invoked from the standard theorem of M. J. Chung and B. Ravikumar (JCSS 39(1), 1989), also cited in I. Parberry, "On the Computational Complexity of Optimal Sorting Network Verification."